

Syllabus

2026 Fall

Course	Embedded Software & Design	Professor	Jeong Hong
Course No	SOC3050	Class No	
Schedule	WED4,5,6,FRI3,4,5(B-101)	Grading Eval.	Relative Evaluation
Other Information			
Profile	- PhD, MIT (Massachusetts Institute of Technology), EECS (Electrical Engineering and Computer Science), USA - MS, EE, KAIST (Korea Advanced Institute of Science and Technology), Korea - BS, EE, SNU (Seoul National University), Korea - Professor, EE, BJTU (Beijing Jiaotong University), China - Professor, EE, POSTECH (Pohang University of Science and Technology), Korea - Professor, EE, KNU (Kyungbook National University), Korea - Visiting Scientist, EE, USC (University of Southern California), USA - Consultant, Dialog Systems, Lucent Bell Labs at Murray Hill, USA		
Course Objectives	This course offers a foundational and hands-on introduction to embedded systems, guiding students from microcontroller architecture to intelligent, real-world applications. Students will learn to program embedded hardware in Assembly and C/C++, interface sensors and actuators, control real-time systems, and deploy lightweight deep learning models for embedded intelligence. The course covers representative processor families including AVR ATmega, Intel-based microcontrollers, and others supported through simulation environments such as SimulIDE and RTOS frameworks. Students will gain experience in both bare-metal programming and operating-system-based development, understanding how processor architecture, memory organization, interrupts, and peripherals shape embedded software design. Through weekly lectures and laboratory sessions, students will build and test embedded applications connected to a computer, culminating in practical systems such as drones, mobile robots, and autonomous devices. By the end of the course, students will have a solid grasp of embedded architectures, low-level programming, real-time control, and the integration of modern AI techniques into resource-constrained hardware.		
Course Description	Embedded Fundamentals AVR architecture and memory organization Instruction set, control flow, and low-level execution Microcontroller design principles Programming Techniques Timers and real-time scheduling Interrupt handling and event-driven control Serial communication (UART, SPI, I²C) Hardware Interfacing LCD modules and display control Sensors (analog/digital) and signal acquisition Motor drivers, PWM, and actuator control Analog/digital I/O management Embedded Intelligence Embeddings and lightweight representation learning Attention mechanisms adapted for constrained devices TinyML deployment on microcontrollers Project Integration IoT system design and communication Automation and control applications		

	AI?enhanced embedded systems (vision, decision?making) Tools & Languages SimulIDE and hardware emulation Atmega development boards Assembly programming C/C++ for embedded software engineering						
Textbooks							
Other Texts and References							
Class Structure	- Lecture - Labs - Project - Exams						
Notes	Course failure: Any one of the following behavior is destined to Failure, - Academic rule: 1/4 (8 times) Absent, <u>unless</u> admitted beforehand by me (not AA) - Any of the following: No Labs, No Project, No Midterm exam, No Final Exam - Cheating in Labs, Homeworks, Projects, and Exams. - Other activity harming the course - Class door will be closed after 5 min of class Also, no class change allowed Course contents may be changed and updated during the semester. Course contents and evaluation criteria may be variable depending on situations during the semester.						
ABEEK							
Grading							
Mid-term	Final exam	Attendance	Assignments	Quiz	Discussion	ETC	Total
30 %	30 %	5 %	30 %	0 %	0 %	5 %	100 %

Syllabus			
Week	Content	Class	Notes
1	Theme	Embedded Systems Overview	
	Class Details	Introduction to computing principles, AVR microcontroller overview, and hardware/software integration concepts.	
	Tests		
2	Theme	AVR Architecture & Programming	
	Class Details	Internal architecture, register operations, instruction set fundamentals, activation logic, and optimization strategies.	
	Tests	Lab	
3	Theme	Control Flow	
	Class Details	Implementation of branching, subroutine calls, and delay loops for timing control.	
	Tests	Lab	
4	Theme	Advanced Instructions	
	Class Details	Arithmetic and logic operations, use of residual connections, and normalization techniques in embedded contexts.	
	Tests	Lab	
5	Theme	AVR I/O Port Programming	
	Class Details	Digital I/O configuration, bit-level manipulation, and port-based device control.	
	Tests		

6	Theme	Deep Learning in Embedded Systems	
	Class Details	Introduction to TinyML, deploying lightweight models, and sensor-driven inference on microcontrollers.	
	Tests		
7	Theme	Midterm Exam	
	Class Details	Assessment covering Weeks 1-6, including theory and practical lab components.	
	Tests	Lab	
8	Theme	Timer Programming	
	Class Details	Timer modes, counters, pulse-width modulation (PWM), and real-time scheduling.	
	Tests		
9	Theme	Interrupt Programming	
	Class Details	Design and implementation of interrupt service routines (ISRs), handling external and internal triggers.	
	Tests	Lab	
10	Theme	Serial Communication	
	Class Details	UART configuration, serial data transmission, and debugging techniques.	
	Tests	Lab	
11	Theme	Display & Input Devices	
	Class Details	Interfacing with LCD displays and matrix keypads for user interaction.	
	Tests	Lab	
12	Theme	Analog Interfacing	
	Class Details	Integration of analog sensors using ADC/DAC modules, signal conditioning, and data acquisition.	
	Tests	Lab	
13	Theme	Actuator Control	
	Class Details	Control of relays, optoisolators, and stepper motors for mechanical actuation.	
	Tests	Lab	
14	Theme	Project Presentation	
	Class Details	Student-led demonstrations of embedded projects, showcasing integration of deep learning and hardware control.	
	Tests		
15	Theme	Final Exam	
	Class Details	Comprehensive evaluation of Weeks 8-13, including system design and interfacing.	
	Tests		
16	Theme	Makeup	
	Class Details		
	Tests		